Flagship Research Direction: A Vision-Language-Action World Model for Fall-Risk-Aware Lower-Limb Exoskeleton Assistance

TL;DR

- The strongest publishable opening is **the first Vision-Language-Action (VLA) model for a human-worn lower-limb exoskeleton**: a VLM-backboned policy that fuses an egocentric/torso camera + on-body proprioception + natural-language assistance/clinical instructions and emits gait-phase-parameterized hip torque, with a learned latent world-model head that predicts the wearer's future margin of stability (MoS) and a fall-risk scalar. No published VLA targets worn assistive lower-limb devices — the entire VLA literature (RT-2 2023, OpenVLA CoRL 2024, π0/π0.5, Octo, NVIDIA GR00T N1, Figure Helix) is manipulation/humanoid — so the novelty lives exactly at the VLA × CV fusion the user wants.
- Concretely: QLoRA fine-tune a PaliGemma/π0-style 3B backbone with a flow-matching action expert (single A100), and add a JEPA-/Dreamer-style latent predictor trained to forecast extrapolated-center-of-mass (XCoM)/MoS and fall-risk from egocentric vision + IMU. Language is *necessary* (clinician assist-as-needed intent, terrain/task priors, risk posture), vision supplies anticipation, action is the torque profile — none alone suffices.
- Headline result that would be CoRL/RSS/NeurIPS-competitive: on real hardware, anticipatory vision+language assistance that (a) **detects perturbations before the wearer's physiological reactive response** and reduces fall/stumble events under slip/trip protocols, while (b) preserving metabolic benefit on par with Molinaro et al.'s unified controller, and (c) generalizing zero-/few-shot to new terrains and stroke gait via language conditioning without retraining the torque controller.

Key Findings (state of the art, 2023–2026)

1. VLA models are exclusively manipulation/humanoid; none control worn lower-limb assistive devices. The flagship families — RT-2 (2023), OpenVLA (CoRL 2024; Kim et al., a 7B model on Llama 2 with a fused DINOv2+SigLIP visual encoder trained on 970k real-world robot demonstrations, beating RT-2-X 55B by 16.5% absolute success across 29 manipulation tasks), Octo, π0 and π0.5 (flow-matching VLAs on a PaliGemma 3B backbone + ~300M action expert), NVIDIA GR00T N1 (humanoid), Figure Helix, Gemini Robotics — all address manipulation or generalist/humanoid robots. "HumanoidExo" and ExoArm-style devices use exoskeletons only as teleoperation data-collection rigs for humanoid manipulation, not as the controlled assistive device. The only language→lower-limb-exo work is speech-command finite-state-machine planning (Guo et al., ICHMS 2024) and upper-limb semantic-intent assistance (arXiv:2508.10378) — neither is a vision-language-action policy. **This is a genuine open gap and the core novelty lever.** (Caveat: confirm against the very newest 2026 preprints before claiming primacy.)

2. CV-for-exoskeletons exists but is narrow and not fused with language or action foundation models. Terrain classification (Laschowski et al., EMBC 2021; gimbal-stabilized MobileNetV2 systems), Exosense vision-centric scene understanding (arXiv:2403.14320), environment-based assistance modulation for a hip exosuit (arXiv:2211.15346), and egocentric optical-flow kinematics prediction (Nature Sci. Data 2023). The anchor group's own CV work (Song, Ivanyuk-Skulskyi, Krieger, Luo, Kang, ICORR 2025) uses human pose estimation to personalize an IMU-based temporal convolutional network (TCN) from only 1–2 gait cycles, reducing kinematics RMSE by 9.7% and 19.9% on stiff-knee gait. All of this is single-modality perception feeding a classifier or estimator — not a unified VLA, and not closed-loop torque.

3. The anchor group has built every component except the fusion.

- *Unified joint-moment control* (Molinaro, Kang, Young, Science Robotics 9(88):eadi8852, 20 Mar 2024): a TCN that "achieved an average root mean square error of 0.142 newton-meters per kilogram across 35 ambulatory conditions without any user-specific calibration" ($R^2 = 0.840 \pm 0.045$). It reduced metabolic cost during level-ground walking by **5.4 ± 5.6% vs. No Exo** and **12.7 ± 2.8% vs. Zero Torque**, and during 5° incline ascent by **10.3 ± 4.4% vs. No Exo** and **17.8 ± 5.1% vs. Zero Torque**; hip-joint positive work fell 29.2 ± 10.9% (level ground).
- *Rapid slip detection* (Peterson, Leestma, Kang, Young, IEEE T-MRB 2025): XGBoost on native hip-exo sensors, **155.06 ms detection at 96.25% accuracy** for early-stance slips (user-independent).
- *Kinematic perturbation detection* (Tagliaferri, Campeggi, Beck, Kang, ICORR 2025): nine-subject cross-validation gives **95.5% accuracy with a 33.1% gait-cycle delay**, outperforming the whole-body-angular-momentum benchmark (cite the published 95.5%/33.1%, not the 98.8%/23.1% pilot).
- *Stroke personalization* (Kang et al., IEEE T-RO 2025): online adaptation improved gait-phase estimation by **40.9% (able-bodied) and 65.9% (stroke)** and cut torque error 32.7% with <1 min of adaptation; also demonstrated cross-hardware transfer to a commercial device.
- *Physics-informed / neuromusculoskeletal sim policies* (Park, Song, Kang, arXiv:2603.04166, 2026 — RL teacher with muscle-synergy prior, sim-to-real $r = 0.82 \pm 0.19$, RMSE 0.03 ± 0.01 Nm/kg; Chiu et al., ICORR 2025 — adversarial imitation, 5.24° RMSE).
- *Conceptual anchor* (Kang, Mitra, Seethapathi, bioRxiv 2026): humans navigate an "environment-dependent risk landscape by mitigating the statistical probability of falling," explained by a *global probabilistic fall risk* rather than local step instability — directly motivating a fall-risk objective for control.

4. World models are mature for legged/humanoid locomotion but unused for human-worn assistance. DreamerV3 (RSSM latent imagination; Hafner et al., Nature 2025), Denoising World Model Learning (RSS 2025 humanoid terrain locomotion), and V-JEPA 2 / V-JEPA 2-AC (Meta, arXiv:2506.09985, released 11 Jun 2025): pretrained on "over 1 million hours of internet-scale video," with a 300M action-conditioned predictor fine-tuned on "less than 62 hours of unlabeled robot videos from the Droid dataset," deployed "zero-shot on Franka arms in two different labs."

These predict robot dynamics — none predict *human* biomechanical stability states for an assistive device.

5. Balance biomechanics give a principled, quantitative target. MoS is the signed distance from the XCoM to the base-of-support boundary; per Hof, Gazendam & Sinke (J. Biomechanics 38:1–8, 2005), "the 'margin of stability' b, the minimum distance from XcoM to the boundaries of the BoS." Beck, Shepherd, Rastogi, Martino, Ting & Sawicki (Science Robotics 8(75):eadf1080, 15 Feb 2023) showed "wearable exoskeletons have the potential to enhance user balance after a disturbance by reacting faster than physiologically possible," with anticipatory ankle torque improving standing balance by **9%** — the strongest motivation for an anticipatory, vision-conditioned predictive controller. For metabolic calibration, Slade, Kochenderfer, Delp & Collins (Nature 610:277–282, 2022) reported real-world personalized assistance "reduced metabolic energy consumption by 23 ± 8% when participants walked on a treadmill at a standard speed of 1.5 m s^{-1}."

Details

Flagship idea: "ExoVLA-WM" — a language-conditioned, vision-grounded, fall-risk-predictive assistance policy

(a) Problem statement & gap. Current lower-limb exoskeleton controllers are reactive and proprioception-only: they estimate biological joint moments or gait phase from on-body IMUs/encoders and apply assistance, but they (i) cannot anticipate terrain or perturbations beyond the body's own sensed deviation, (ii) cannot be re-tasked by natural language, and (iii) optimize metabolic cost or moment-tracking without an explicit, predictive model of fall risk. We propose a single VLA policy that ingests an egocentric/torso RGB(-D) camera, on-body proprioception (IMU + exo encoders), and a natural-language instruction, and outputs gait-phase-parameterized hip assistance torque — while a coupled latent world-model head predicts the future XCoM/MoS trajectory and a scalar fall-risk probability that gates assistance aggressiveness. The gap filled: vision-language-action control for a *human-worn* device where the "agent" being acted upon is a human body with its own intent and fall-risk landscape.

(b) Novelty vs. anchors & literature. Differentiates from Molinaro (no vision, no language, reactive moment estimation), Song ICORR 2025 (vision only personalizes an estimator offline; no language, no closed-loop torque, no world model), Tagliaferri/Peterson (detection only; no anticipatory vision, no language), and the Kang stroke T-RO work (proprioceptive online adaptation, no vision/language). Vs. the broader VLA field: first application to worn assistive lower-limb robotics; introduces a *biomechanical* world-model head (XCoM/MoS/fall-risk) rather than pixel/latent-state reconstruction. It operationalizes the Kang–Mitra–Seethapathi fall-risk-aware adaptation thesis as an explicit control objective.

(c) System/architecture.

- *Vision encoder:* SigLIP/PaliGemma vision tower (or DINOv2, as in OpenVLA) on a torso/egocentric camera; optionally a lightweight depth/elevation feature (RealSense) for terrain geometry. Pretrained features frozen or LoRA-adapted.

- *Language interface:* the VLM backbone (Gemma-2B-class). Language carries (1) clinician assist-as-needed intent ("provide gentle propulsion, prioritize stability on the paretic side"), (2) task/terrain priors ("descending stairs," "icy ramp"), (3) risk posture ("conservative — patient is fall-prone"). Language is the natural channel for clinician-in-the-loop re-tasking and compositional generalization to unseen terrain/task descriptions.
- *Action space:* gait-phase-parameterized hip torque (peak magnitude, end timing, duration per the Zhang/Ding human-in-the-loop parameterization) plus a continuous bilateral torque chunk, generated by a π0-style flow-matching action expert (~300M params) conditioned on proprioception tokens. Action chunking of ~50 steps with asynchronous re-planning at a ~100–200 Hz control loop. arxiv
- *World-model / predictive head:* a JEPA-/Dreamer-style latent predictor that, from egocentric vision + proprioception history, forecasts future XCoM/MoS and a fall-risk scalar over a ~0.5–1 s horizon. This both (i) feeds an anticipatory feature to the action expert and (ii) supplies a control-barrier-function-style fall-risk gate on torque aggressiveness.
- *Biomechanics constraints:* train/regularize with a neuromusculoskeletal simulator in the loop (OpenSim/MuJoCo musculoskeletal, as in Park et al. 2026 and Chiu et al. 2025) so assistance respects muscle-synergy priors and joint limits; reward shaped by EMG reduction + metabolic surrogate + MoS preservation.

(d) Dataset & hardware plan. Using the group's hip exoskeleton + Humotech/treadmill perturbation platform:

- *Collect:* time-synced egocentric/torso video, IMU+encoder proprioception, motion-capture ground truth (for XCoM/MoS labels), EMG, and indirect calorimetry, across level ground, ramps, stairs, and obstacle/terrain transitions; plus controlled slip/trip and AP/ML treadmill-belt perturbations. Pair each trial with natural-language captions/instructions (auto-generated from terrain/task labels + clinician-style templates) to build the language channel and enable a release as a benchmark.
- *Baselines:* (1) Zero-torque; (2) Molinaro-style unified TCN moment controller (proprioception only); (3) vision-augmented classifier→assistance (Song/terrain-classification style); (4) ablations removing language, removing vision, removing the world-model head.
- *Perturbation protocol:* early-/late-stance slips of varied magnitude (Peterson protocol) and AP/ML treadmill perturbations; measure detection lead time relative to physiological reactive-moment onset (Beck framing).
- *Metrics:* metabolic cost (W/kg, %), MoS preservation, fall/stumble event count and recovery success, perturbation detection lead time (ms vs. physiological onset), joint-moment tracking RMSE (Nm/kg), EMG reduction, gait-phase error; language generalization (held-out instructions/terrains); stroke-gait transfer (few-shot).

(e) Evaluation & headline result. Competitive claim: a *single* vision+language+action policy that (i) anticipates perturbations and reduces fall/stumble events vs. all baselines, reacting before physiological reactive-moment onset (Beck et al. motivation); (ii) preserves metabolic benefit on par with the unified controller on level ground/ramps (target: within noise of the ~5–

13% level-ground and ~10–18% incline reductions Molinaro reported); and (iii) generalizes zero-/few-shot to new terrains and to stroke gait via language conditioning, without retraining the torque controller. The figure-1 story: "re-task a worn exoskeleton with a sentence, and it stays safe because it predicts your fall risk from what it sees."

(f) Anticipated reviewer objections & preemptions.

- *"Is language necessary or decorative?"* Ablate: show language conditioning yields measurable zero-shot terrain/stroke transfer and clinician-controllable assist-as-needed behavior that proprioception-only baselines cannot. Frame language as the clinician/HITL interface.
- *"VLA is overkill for torque control."* Position the contribution as the predictive fall-risk world model + cross-task/terrain generalization, not raw torque tracking; show generalization a TCN cannot match.
- *"Safety of a large model in the loop."* Dual-system design: a fast low-level safety filter / CBF gated by predicted MoS runs at high rate; the VLA re-plans chunks asynchronously. Report the latency budget and fallback policy.
- *"N is small / single lab."* Standard for hardware exo papers; emphasize within-subject perturbation statistics, real-robot deployment, and a released dataset (NeurIPS D&B angle).
- *"Compute / reproducibility."* QLoRA fine-tuning on a single A100; prototype on T4/A10 with a frozen backbone first.

(g) Why fusion is necessary, not incidental. Vision supplies *anticipation* (terrain/perturbation cues before bodily deviation) that proprioception cannot; language supplies *intent and compositional re-tasking* (clinician assist-as-needed, risk posture, unseen-terrain priors) that vision cannot encode; action (torque) is the only thing that changes the wearer's biomechanics. The fall-risk world model only becomes controllable when all three are present — vision to predict, language to set the risk/intent trade-off, action to intervene. Remove any modality and either anticipation, controllability, or efficacy collapses.

Compute & prototyping

Single A100 (80GB) for QLoRA fine-tuning of a 3B-class VLM + flow-matching head; few A10/T4 GPUs for the frozen-backbone prototype and the world-model head (small JEPA/RSSM). Prototype offline on collected data (open-loop torque prediction + MoS forecasting) before closed-loop hardware deployment.

Backup ideas (sketched)

- **B1 — "Biomechanical World Model" (NeurIPS/ICLR, simpler):** drop language; train a Dreamer/V-JEPA-style latent world model that predicts XCoM/MoS and fall risk from egocentric vision + IMU, used for model-predictive anticipatory assistance. Cleaner ML story, less HRI risk; weaker on the "fusion" mandate.
- **B2 — "Language-personalized assist-as-needed for stroke" (T-RO/TNSRE/ICORR):** clinician natural-language instructions → assistance-parameter adaptation on top of the

Kang stroke online-adaptation framework, with pose-estimation vision for asymmetry monitoring. Strong clinical translation; lighter on novelty for ML venues.

Recommendations

1. **Stage 0 (weeks):** Offline proof-of-concept on existing logged data — train the world-model head to forecast MoS/fall-risk from vision+IMU; show it beats proprioception-only and the angular-momentum benchmark (Tagliaferri's 95.5%/33.1%) on perturbation anticipation. Proceed if it yields predictive lead time *before* physiological onset at useful accuracy.
2. **Stage 1:** QLoRA fine-tune the VLA on collected trials with auto-generated language; validate open-loop torque + chunked action prediction against the unified-controller baseline. Proceed if moment-tracking RMSE and the metabolic surrogate match the TCN (≈0.142 Nm/kg) within noise.
3. **Stage 2:** Closed-loop hardware with the safety filter; run slip/trip + treadmill perturbation protocols; measure fall/stumble reduction and detection lead time. Threshold for a top-tier submission: statistically significant fall-event reduction AND preserved metabolic benefit AND nonzero language/terrain generalization.
4. **Venue routing:** Lead for **CoRL** (real-robot learning, language-conditioned) or **RSS**; if the dataset is strong, a parallel **NeurIPS Datasets & Benchmarks** submission (first vision-language-exoskeleton dataset). Clinical-translation results → **Science Robotics / IEEE T-RO / TNSRE**. Calibrate to CoRL 2025 norms (real-hardware demos, ablation-heavy, released code).
5. **De-risk the negative claim:** before asserting "first VLA for worn exoskeletons," do a final manual scan of late-2025/2026 arXiv/CoRL/ICRA — the VLA field moves fast.

Caveats

- The "no prior VLA for worn lower-limb exo" claim is based on current surveys and targeted search; verify against the newest preprints before claiming primacy.
- Hardware exo studies are small-N and single-lab; statistical power for fall-event reduction must be designed carefully (within-subject design, many perturbation trials).
- Large-model in-the-loop latency is a real risk; the dual-system safety architecture is essential and must be validated empirically.
- Metabolic and EMG measurements are noisy; use established ML surrogates (e.g., Slade-style real-time estimation) and adequate trial counts.
- Tagliaferri figures differ between the pilot preprint (98.8%/23.1%, 5 subjects) and the published ICORR 2025 version (95.5%/33.1%, 9 subjects); cite the published numbers.